

Questions and Answers with Bram Cohen, Class of 1993



Bram Cohen, Class of 1993, famously created the peer-to-peer BitTorrent protocol in 2005 in his early thirties and the Chia cryptocurrency, the first enterprise-grade digital money, in 2017 in his forties. As a Stuyvesant student, he participated in Math Team and qualified for the USA Math Olympiad. During his high school years, Cohen also attended the Hampshire College Summer Studies in Mathematics program. Although he attended SUNY Buffalo for one year, he ultimately left to work for several startup companies, such as Valve. Then, he founded BitTorrent, which grew to fame, with over 150 million global users, due to its ability to quickly share large music and movie files online. Later, cryptocurrency startup TRON acquired BitTorrent, and Cohen left to start the groundbreaking Chia Network, a proof of space-time cryptocurrency. Chia launched in May 2021 and uses about 10,000 times less energy than what is usually needed to mine proof-of-work-based cryptocurrencies such as Bitcoin. Its self-reported market cap is currently at around \$244 million and has driven hard drive disk sales soar to record levels, increasing by 141 percent.

Bram Cohen's answers are taken from an interview conducted by Josephine Lee and have been edited for clarity.

1) How did you get interested in math and computer science? Did you always envision being an entrepreneur?

Reading books by Martin Gardner, when I was a kid, actually was a lot of my introduction to math. There were also collections of his mathematical games and columns from *Scientific American* that I read rather obsessively. Then I was part of Math Team starting in junior high. As far as being an entrepreneur, I didn't decide, "Oh, I want to be an entrepreneur." I decided, "Hey, there's this project I want to work on." And that turned into being an entrepreneur. So I guess I was always an entrepreneur.

2) What do you remember most about your experience at Stuyvesant?

Most of my experience at Stuyvesant was really centered around being on Math Team, which was run by Mr. Geller at the time. Mostly, I just remember that Math Team was actually fun.

3) How did you get BitTorrent started? What was the process in creating BitTorrent and getting people to use it? How were you able to get your technology widely adopted?

Oh, well, I just kind of wrote it. I had worked at a bunch of startups doing protocol stuff. And from those experiences, I had thoughts of something that would be useful. So I built the thing that was useful, and then I released it, and it was free, and people found it helpful. So they used it.

4) What was the process in creating Chia? Are there any difficulties in expanding, creating, and distributing Chia that you hadn't experienced before?

Chia is a much bigger project than BitTorrent in terms of scope, getting widespread mass adoption and enabling real commerce in the world. I come into Chia having a much, much better idea of what I'm doing. I also have more of a network, so that's been better. But it's also been a much larger project to even get started in the first place.

5) What advice do you have for high school students who want to have a career in CS and/or to be an entrepreneur in CS? Are there particular skills one should focus on?

For being an entrepreneur, I would say the big mistake many people make is jumping directly into being an entrepreneur with no work experience. You're way, way better off working at startups, where you're one of the first ten employees or something like that. Because then, you get some idea what's going on in it. You build a network, and you have some idea what to expect rather than just flailing completely on your own. It's better to be paid to gain an audience than just to give away money. And then, after you have some experience, then you can try being an entrepreneur.

6) What lessons have you learned by going through the process of developing, marketing, and selling a company that might be helpful for high school students who want to be entrepreneurs?

Chia is very tech-heavy and has a firm focus there since we're trying to be the next chapter of digital money. The weakness that most companies have is in the product, not so much in engineering. People have reasonably good ideas on how to do engineering, for the most part, these days. But the whole way in which companies approach, "Okay, what are we doing? Why are we doing it?" tends to be very slapdash and haphazard. Partially this isn't done well because there aren't obvious guidelines about how to get better at it, but as an entrepreneur, you should try to focus on really understanding what things are like from a product standpoint.